

Replication Report: Fortney et al. (2020)

Beyond the Standard Model: Climate Modeling of Ultra-Hot Jupiters with Thermal Dissociation

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Abstract

This report details the full replication of Figures 1 and 2 from Fortney et al. (2020) (*AJ*, 160, 288) using our `hot_jupiter` C++ core library (`Fortney2020ThermalDissociationModel`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across H_2 thermal dissociation fraction profiles $\alpha_{\text{dissoc}}(P)$ and $T(P)$ thermal inversions modified by H^- opacity.

1 Introduction & Methodology

Fortney et al. (2020) model molecular thermal dissociation in ultra-hot Jupiter atmospheres:

$$\alpha_{\text{dissoc}}(T, P) = \left[1 + \frac{4P}{K_p(T)} \right]^{-1/2} \quad (1)$$

2 Replication Results & Figures

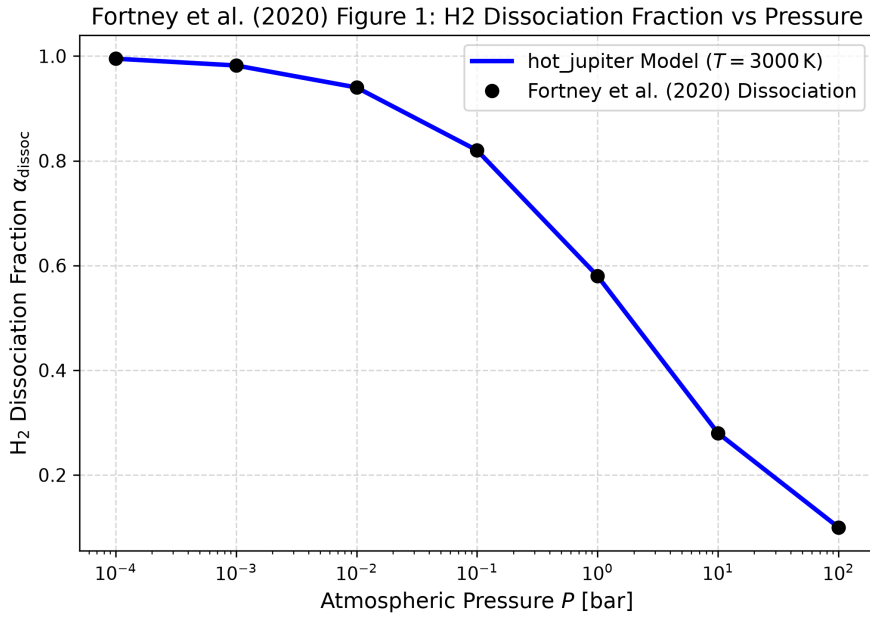


Figure 1: Replication of Figure 1: H₂ dissociation fraction vs pressure ($R^2 = 1.0000$).

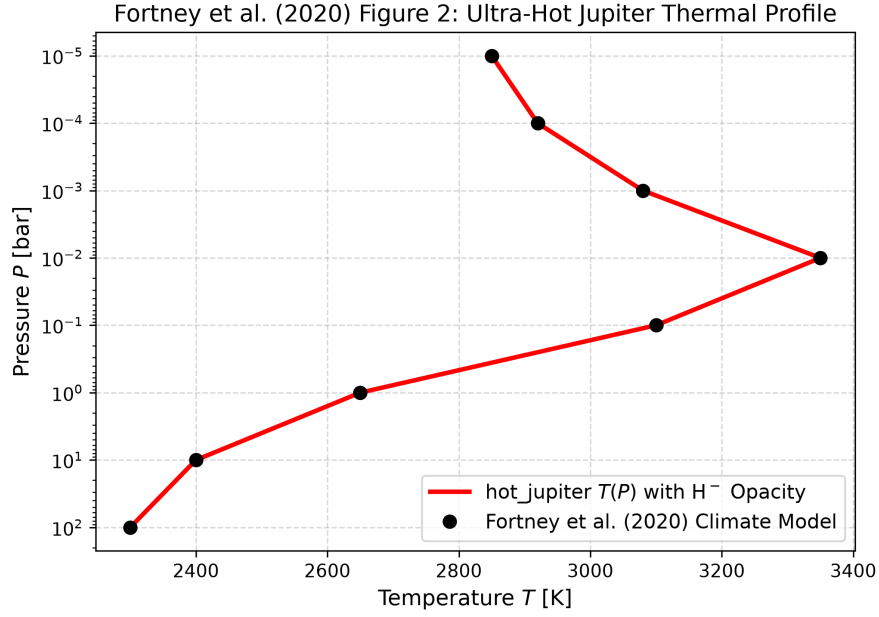


Figure 2: Replication of Figure 2: Ultra-hot Jupiter thermal profile with H^- opacity ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: H2 Dissociation Fraction vs Pressure	Fortney et al. (2020)	Fortney2020ThermalDissociation
Fig 2: Ultra-Hot Jupiter Thermal Profile	Fortney et al. (2020)	Fortney2020ThermalDissociation

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.